GENERALIZED HERMITE–BIRKHOFF INTERPOLATION OF ARBITRARY ORDER IN ANY DIMENSION VIA SINGULAR-WEIGHT MOVING LEAST SQUARES

DAVID FIGUEIRAS GARCIA

ABSTRACT. [**TODO:** polish last.] Singular-weight interpolating moving least squares (SWIMLS) reproduces function values at scattered nodes. We study the method when each node carries an arbitrary finite set of linear differential data $q(D)f(x_i)$. We prove that the interpolant is C^∞, give an exact representation of the error in every datum, and deduce that a datum is reproduced exactly if (i) every derivative of its symbol q lies in the span of the symbols prescribed at that node and (ii) $\Delta^{\alpha/2}q = 0$, where α is the (even) order of the weight singularity. Condition (ii) cannot be dropped, whereas condition (i) is needed or not depending on how the weights of the other nodes vary near x_i, which we make precise. For classical Hermite data of order k, $\alpha > k$ suffices, which proves a sharpened form of a conjecture of Komargodski and Levin; we also give exact counterexamples, in one and two dimensions, to their claim that derivatives prescribed at points without function values are interpolated, and a completion procedure that makes such data exact.

1. INTRODUCTION

[**TODO:** Context: MLS (McLain, Lancaster–Šalkauskas), interpolating MLS with singular weights, Shepard; Hermite data in meshless methods; multivariate Hermite–Birkhoff interpolation and poisedness (Gasca–Sauer); Komargodski–Levin. Outline.]
Contributions.

(1) A structure theorem for SWIMLS near a node (Theorem 4.1): the interpolant is C^∞, and an exact representation of the error in any differential datum (Theorem 4.3).
(2) A sufficient condition for exact reproduction of a datum $q(D)f(x_i)$: closure of the symbol under differentiation and $\Delta^{\alpha/2}q = 0$ (Corollary 4.4). The second condition cannot be dropped: it is necessary at leading order (Proposition 4.6).
(3) The role of the first condition: it is not needed when the minimiser of the "frozen" problem at the other nodes does not move near x_i (Theorem 4.8), e.g. for a single other node or locally proportional weights; in general its failure produces the explicit first-order error of Proposition 4.11.
(4) For classical Hermite data of order k: all data are reproduced when $\alpha > k$, and this is sharp (Corollaries 4.5 and 4.7); when $\alpha \le k$ the error is the linear functional $q \mapsto \left[\Delta^{\alpha/2}q, \tau\right]_{x_i}$ of the datum. This proves, for $k \ge 3$ in a sharper form, the conjecture of [1, §3] that $\alpha = 2k$ suffices.
(5) Exact counterexamples in one and two dimensions to the extension claimed in [1, §3] (Propositions 5.1 and 5.2), and a completion procedure that makes data with gaps exact (Proposition 5.3).
(6) A numerically stable evaluation of SWIMLS, including the closed-form limit at the nodes, and numerical experiments in one, two and three dimensions.

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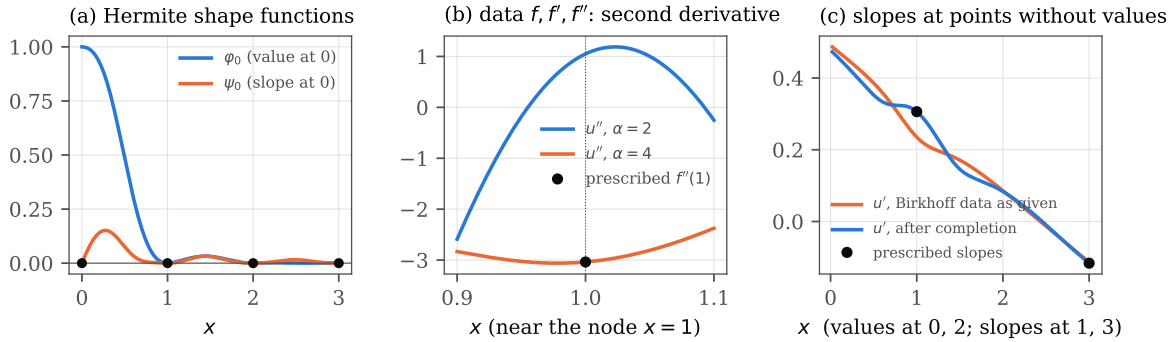


FIGURE 1. The one-dimensional example. (a) Shape functions for the value and the slope at $x = 0$ (nodes $0, 1, 2, 3$, $m = 2$, $\alpha = 2$). (b) Data f, f', f'' at the nodes: u'' near the node $x = 1$ for $\alpha = 2$ and $\alpha = 4$; only $\alpha = 4$ reproduces $f''(1)$ (Corollary 4.5). (c) Values at $0, 2$ and slopes at $1, 3$: the slopes are not reproduced (Proposition 5.1) until the data are completed (Proposition 5.3).

2. A one-dimensional example

[**TODO:** About one page: nodes $0, 1, 2, 3$, values and slopes, $\alpha = 2$, local quadratics; shape functions φ_0, ψ_0 with the Hermite cardinal property; second derivatives need $\alpha = 4$; slopes at points without values are not reproduced and completion repairs them. Figure 1.]

3. Setting and notation

3.1. **Polynomials, symbols and the Fischer pairing.** For $d \geq 1$ and $m \geq 0$ let $\Pi_m = \Pi_m(\mathbb{R}^d)$ be the real polynomials of total degree at most m. Multi-indices are $\beta \in \mathbb{N}_0^d$, with $|\beta| = \sum_j \beta_j$, $\beta! = \prod_j \beta_j!$, $D^\beta = \partial^{|\beta|}/\partial x_1^{\beta_1} \cdots \partial x_d^{\beta_d}$. A polynomial $q(\xi) = \sum_\beta q_\beta \xi^\beta$, called a *symbol*, defines the differential operator $q(D) = \sum_\beta q_\beta D^\beta$. We write $\partial^\gamma q$ for derivatives of a symbol with respect to ξ, and $\Delta q = \sum_j \partial_j^2 q$.

For a symbol q, a smooth function g and a point z we set

$$[q, g]_z := \big(q(D)g\big)(z), \qquad [q, g] := [q, g]_0. \tag{1}$$

For polynomials q, p, $[q, p] = \sum_\beta \beta!\, q_\beta p_\beta$ is the *Fischer inner product*; it is positive definite on Π_m. Four elementary facts will be used.

Lemma 3.1 (Fischer adjointness)**.** *For symbols q and polynomials p, $[q, z_j p] = [\partial_j q, p]$. Consequently $[q, |z|^{2\ell} p] = [\Delta^\ell q, p]$ for every $\ell \in \mathbb{N}_0$, where $|z|^2 = \sum_j z_j^2$.*

Proof. With e_j the j-th unit multi-index, $[q, z_j p] = \sum_\kappa \kappa!\, q_\kappa p_{\kappa - e_j} = \sum_\lambda (\lambda + e_j)!\, q_{\lambda + e_j} p_\lambda = \sum_\lambda \lambda!\, (\lambda_j + 1) q_{\lambda + e_j} p_\lambda = [\partial_j q, p]$. Multiplication by $|z|^2 = \sum_j z_j \cdot z_j$ is therefore adjoint to $\sum_j \partial_j^2 = \Delta$, and the second statement follows by induction. □

Lemma 3.2 (Taylor expansion of operators)**.** *Let $A = (A_1, \ldots, A_d)$ and $B = (B_1, \ldots, B_d)$ be commuting operators. For every symbol q, $q(A + B) = \sum_\gamma \frac{1}{\gamma!} A^\gamma (\partial^\gamma q)(B)$ (a finite sum).*

Proof. This is Taylor's formula $q(a + b) = \sum_\gamma \frac{a^\gamma}{\gamma!} (\partial^\gamma q)(b)$ for the polynomial q, an identity in the commutative polynomial ring in a, b. □

Lemma 3.3. *Let $\ell \in \mathbb{N}_0$ and $\beta \in \mathbb{N}_0^d$. Then $\Delta^\ell \xi^\beta = 0$ if and only if $\sum_j \lfloor \beta_j / 2 \rfloor < \ell$.*

Proof. By the multinomial formula, $\Delta^\ell \xi^\beta = \sum_{|\kappa|=\ell} \frac{\ell!}{\kappa!}\, \partial^{2\kappa}\xi^\beta$, and $\partial^{2\kappa}\xi^\beta = \frac{\beta!}{(\beta-2\kappa)!}\xi^{\beta-2\kappa}$ if $2\kappa \le \beta$ and 0 otherwise. Distinct κ give distinct monomials and all coefficients are positive, so $\Delta^\ell \xi^\beta \neq 0$ if and only if some κ with $|\kappa| = \ell$ satisfies $2\kappa \le \beta$, i.e. $\sum_j \lfloor \beta_j/2 \rfloor \ge \ell$. □

Lemma 3.4. *Let $\alpha \in 2\mathbb{N}$. Then $\sum_{|\gamma|=\alpha} \frac{1}{\gamma!}\big(D^\gamma |x|^\alpha\big)(0)\, \partial^\gamma = \Delta^{\alpha/2}$ as operators on symbols.*

Proof. $|x|^\alpha = \sum_{|\kappa|=\alpha/2} \frac{(\alpha/2)!}{\kappa!} x^{2\kappa}$, and $(D^\gamma x^{2\kappa})(0) = (2\kappa)!$ if $\gamma = 2\kappa$ and 0 otherwise. Hence the left side equals $\sum_{|\kappa|=\alpha/2} \frac{(\alpha/2)!}{\kappa!}\partial^{2\kappa} = \big(\sum_j \partial_j^2\big)^{\alpha/2}$. □

3.2. **Data.** Let $\Omega \subset \mathbb{R}^d$ be open and $X = \{x_1, \dots, x_K\} \subset \Omega$ distinct nodes. Node x_i carries symbols $q_{i,1}, \dots, q_{i,n_i}$ of degree at most m and values $y_{i,1}, \dots, y_{i,n_i} \in \mathbb{R}$, i.e. the interpolation conditions

$$\lambda_{i,s}(f) := \big[q_{i,s}, f\big]_{x_i} = \big(q_{i,s}(D)f\big)(x_i) = y_{i,s}, \qquad s = 1, \dots, n_i,\ i = 1, \dots, K. \tag{2}$$

We write $Q_i = \operatorname{span}\{q_{i,1}, \dots, q_{i,n_i}\}$ for the space of symbols at x_i. Every $q = \sum_s c_s q_{i,s} \in Q_i$ has a prescribed value $y_q := \sum_s c_s y_{i,s}$; we say that the datum q is *reproduced* by a function u if $\big[q, u\big]_{x_i} = y_q$. Classical Hermite data of order k correspond to $Q_i = \operatorname{span}\{\xi^\beta : |\beta| \le k\}$; any other choice will be called Hermite–Birkhoff data. A space of symbols is *D-invariant* if it is closed under differentiation.

Assumption 3.5. (A1) $\alpha \in 2\mathbb{N}$ is even, and $w_i(x) = \rho_i(x)\,|x - x_i|^{-\alpha}$ with $\rho_i \in C^\infty(\Omega)$, $\rho_i > 0$.
(A2) For every i the symbols $q_{i,1}, \dots, q_{i,n_i}$ are linearly independent.
(A3) (Unisolvency) If $p \in \Pi_m$ and $\lambda_{i,s}(p) = 0$ for all i, s, then $p = 0$.

Since $\big[\cdot,\cdot\big]_{x_i}$ is non-degenerate on Π_m (it is the Fischer pairing after translation) and $\deg q_{i,s} \le m$, (A2) is equivalent to the linear independence of the functionals $\lambda_{i,1}, \dots, \lambda_{i,n_i}$ on Π_m.

Definition 3.6 (SWIMLS)**.** For $x \in \Omega \setminus X$ let

$$J_x(p) = \sum_{i=1}^{K} w_i(x) \sum_{s=1}^{n_i} \big(\lambda_{i,s}(p) - y_{i,s}\big)^2, \qquad p_x := \operatorname*{arg\,min}_{p \in \Pi_m} J_x(p), \tag{3}$$

and define the interpolant $u(x) := p_x(x)$.

By (A1) and (A3), J_x is a strictly convex quadratic on Π_m for every $x \notin X$, so p_x is unique; it depends smoothly on $x \in \Omega \setminus X$ because the coefficients of J_x do and its Hessian is positive definite. Hence $u \in C^\infty(\Omega \setminus X)$. The behaviour at the nodes is the subject of the next section.

Remark 3.7. Positive smooth factors for the individual functionals of a node can be included: the first block of the Jacobian in the proof of Theorem 4.1 becomes $2L^*W(x)L$, still invertible, and nothing else changes. If some functionals at x_i are *not* given the singular factor, they must be moved from Q_i to the regular part B_x of the proof, and Q_i, N_i refer to the singular ones only.

4. Main results

Throughout this section we fix a node x_i and write

$$N_i = \{p \in \Pi_m : \lambda_{i,s}(p) = 0,\ s = 1, \dots, n_i\} = \{p \in \Pi_m : \big[q, p\big]_{x_i} = 0\ \forall q \in Q_i\}$$

for the polynomials that the data at x_i do not see.

4.1. **Structure near a node.**

Theorem 4.1 (Structure near a node)**.** *Let Assumption 3.5 hold, fix an inner product on Π_m and let $R = N_i^{\perp}$. There are an open neighbourhood $U \subset \Omega$ of x_i with $U \cap X = \{x_i\}$, a polynomial $p_i^* \in R$ with $\lambda_{i,s}(p_i^*) = y_{i,s}$ for all s, and C^∞ maps $\nu : U \to N_i$ and $\sigma : U \to R$ such that*

$$p_x = p_i^* + \nu(x) + |x - x_i|^\alpha \, \sigma(x) \qquad \text{for all } x \in U \setminus \{x_i\}. \tag{4}$$

More precisely, $\nu(x) = n(x, |x-x_i|^\alpha)$ and $\sigma(x) = s(x, |x-x_i|^\alpha)$ for C^∞ maps n, s defined near $(x_i, 0)$, and $Z(x) := n(x, 0)$ is the frozen minimiser

$$Z(x) = \operatorname*{arg\,min}_{n \in N_i} \; B_x(p_i^* + n), \qquad B_x(p) := \sum_{j \neq i} \frac{w_j(x)}{\rho_i(x)} \sum_{s=1}^{n_j} \big(\lambda_{j,s}(p) - y_{j,s}\big)^2. \tag{5}$$

Applying this at every node, u extends to a function in $C^\infty(\Omega)$, with node value $u(x_i) = \big(p_i^* + Z(x_i)\big)(x_i)$.

Proof. Write $\Lambda_i : \Pi_m \to \mathbb{R}^{n_i}$, $p \mapsto (\lambda_{i,s}(p))_s$, and Λ_j likewise. By (A2) Λ_i is surjective with kernel N_i, so $L := \Lambda_i|_R : R \to \mathbb{R}^{n_i}$ is bijective; put $p_i^* := L^{-1} y_i \in R$. Every $p \in \Pi_m$ is uniquely $p = p_i^* + r + n$ with $r \in R$, $n \in N_i$, and then $\Lambda_i p - y_i = Lr$.

Let U be an open neighbourhood of x_i containing no other node. For $x \in U \setminus \{x_i\}$ divide J_x by $w_i(x) > 0$:

$$\frac{J_x(p)}{w_i(x)} = |Lr|^2 + t(x)\, B_x(p), \qquad t(x) := |x - x_i|^\alpha,$$

with B_x as in (5). Since α is even, $t(x) = \big(\sum_k (x_k - x_{i,k})^2\big)^{\alpha/2}$ is a polynomial in x; and $x \mapsto B_x$ is a C^∞ family of convex quadratics on U, because $\rho_i > 0$ and the w_j, $j \neq i$, are smooth on U. A polynomial $p = p_i^* + r + n$ minimises J_x if and only if the gradient vanishes (the problem is convex), that is

$$2L^*Lr + t(x) \nabla_R B_x(p) = 0, \qquad t(x) \nabla_N B_x(p) = 0,$$

where ∇_R, ∇_N denote the components of the gradient in R and N_i and L^* is the adjoint of L. Since $t(x) > 0$ for $x \neq x_i$ we may substitute $r = t(x)\, s$ and divide by $t(x)$, obtaining $F(x, t(x); s, n) = 0$ for the map $F : U \times \mathbb{R} \times R \times N_i \to R \times N_i$,

$$F(x, t; s, n) := \Big(2L^*Ls + \nabla_R B_x(p_i^* + ts + n),\; \nabla_N B_x(p_i^* + ts + n)\Big),$$

which is C^∞ in all variables, including t near 0.

The point $(x_i, 0)$. Here $F = 0$ reads $\nabla_N B_{x_i}(p_i^* + n) = 0$ and $s = -(2L^*L)^{-1} \nabla_R B_{x_i}(p_i^* + n)$. The restriction of B_x to the affine space $p_i^* + N_i$ is strictly convex for every $x \in U$: its Hessian is $2 \sum_{j \neq i} \frac{w_j(x)}{\rho_i(x)} \Lambda_j^* \Lambda_j$ restricted to N_i, with $w_j(x) > 0$ since $x \neq x_j$; and if $n \in N_i$ satisfies $\Lambda_j n = 0$ for all $j \neq i$ then all data functionals vanish on n, so $n = 0$ by (A3). Hence there is exactly one solution (s_0, n_0), and $n_0 = Z(x_i)$; more generally, for every $x \in U$ the equation $F(x, 0; s, n) = 0$ has the unique solution $n = Z(x)$, $s = -(2L^*L)^{-1} \nabla_R B_x(p_i^* + Z(x))$.

Implicit function theorem. At $(x_i, 0; s_0, n_0)$ the derivative of F with respect to (s, n) is block triangular,

$$\frac{\partial F}{\partial (s, n)} = \begin{pmatrix} 2L^*L & \big(\nabla^2 B_{x_i}\big)_{RN} \\ 0 & \big(\nabla^2 B_{x_i}\big)_{NN} \end{pmatrix},$$

because every derivative with respect to s of a B-term carries the factor $t = 0$. Both diagonal blocks are invertible (L is injective; the second block is the Hessian above). By the implicit function theorem there are a neighbourhood $U' \times (-\tau_0, \tau_0)$ of $(x_i, 0)$ and a C^∞ map $(x, t) \mapsto (s(x,t), n(x,t))$ with $F(x, t; s(x,t), n(x,t)) = 0$; by the uniqueness just shown, $n(x, 0) = Z(x)$.

Shrinking U to $U' \cap \{t(x) < \tau_0\}$, for $x \in U \setminus \{x_i\}$ the polynomial $p_i^* + t(x)s(x,t(x)) + n(x,t(x))$ satisfies the gradient equations of the strictly convex function J_x, hence it equals p_x (only uniqueness of the critical point of J_x is used). Setting $\nu(x) := n(x,t(x)) \in N_i$ and $\sigma(x) := s(x,t(x)) \in R$ gives (4); both maps are C^∞ because t is a polynomial. Finally $u(x) = p_x(x)$ is, by (4), the restriction to $U \setminus \{x_i\}$ of the C^∞ function $x \mapsto (p_i^* + \nu(x) + t(x)\sigma(x))(x)$, whose value at x_i is $(p_i^* + Z(x_i))(x_i)$. □

Remark 4.2. The decomposition (4) is canonical up to the choice of the complement R of N_i; the node value $u(x_i)$ and all errors below do not depend on it. Evenness of α is used twice: to make $t(x) = |x - x_i|^\alpha$ smooth at x_i (Theorem 4.1) and to make it a polynomial, so that Lemmas 3.1 and 3.4 apply (Theorems 4.3 and 4.8). If the ρ_i are real analytic, the analytic implicit function theorem shows that u is real analytic. [**TODO:** odd α: smoothness at the nodes is limited; give an example or drop.]

4.2. **An exact error representation.**

Theorem 4.3 (Error representation)**.** *With the notation of Theorem 4.1, let $\tau(z) := \sigma(z)(z)$ for $z \in U$ (a C^∞ scalar function). For every symbol q,*

$$[q,u]_{x_i} = [q,p_i^*]_{x_i} + \sum_\gamma \frac{1}{\gamma!} D^\gamma \Big[x \mapsto [\partial^\gamma q, \nu(x)]_{x_i}\Big](x_i) + [\Delta^{\alpha/2} q, \tau]_{x_i}. \tag{6}$$

The $\gamma = 0$ term of the middle sum is $[q, \nu(x_i)]_{x_i}$.

Proof. On $U \times \mathbb{R}^d$ consider the C^∞ function $\Phi(x,y) = p_i^*(y) + \nu(x)(y) + t(x)\,\sigma(x)(y) =: \Phi_1 + \Phi_2 + \Phi_3$, which is a polynomial in y and satisfies $u(z) = \Phi(z,z)$ for all $z \in U$, including $z = x_i$ by the definition of the extension. By the chain rule along the diagonal, $D_z^\beta[\Phi(z,z)] = \big[(D_x + D_y)^\beta \Phi\big](z,z)$, hence $[q,u]_{x_i} = \big[q(D_x + D_y)\Phi\big](x_i,x_i)$. We treat the three parts.

Φ_1 does not depend on x, so it contributes $(q(D)p_i^*)(x_i) = [q,p_i^*]_{x_i}$.

For Φ_2, Lemma 3.2 with $A = D_x$, $B = D_y$ gives $q(D_x + D_y)\Phi_2 = \sum_\gamma \frac{1}{\gamma!} D_x^\gamma\, (\partial^\gamma q)(D_y)\Phi_2$. Setting $y = x_i$ commutes with D_x, and $\big((\partial^\gamma q)(D_y)\Phi_2\big)(x,x_i) = [\partial^\gamma q, \nu(x)]_{x_i}$; setting $x = x_i$ gives the middle term of (6).

For Φ_3 we use the diagonal again: its contribution is $[q, t\,\tau]_{x_i} = \big(q(D)[\,|z - x_i|^\alpha \tau(z)]\big)(x_i)$. Translate x_i to 0 and let $T\tau$ be the Taylor polynomial of τ at 0 of degree $\deg q$. Since $|z|^\alpha(\tau - T\tau)$ vanishes at 0 to order larger than $\deg q$, it is annihilated by $q(D)$ at 0, and by Lemma 3.1

$$[q, |z|^\alpha \tau] = [q, |z|^\alpha T\tau] = [\Delta^{\alpha/2} q, T\tau] = [\Delta^{\alpha/2} q, \tau],$$

the last equality because $\deg \Delta^{\alpha/2} q \le \deg q - \alpha \le \deg T\tau$. □

4.3. **Exact reproduction.**

Corollary 4.4 (Sufficient condition)**.** *Let Assumption 3.5 hold and let $q \in Q_i$. If*

(i) $\partial^\gamma q \in Q_i$ for every multi-index γ, and

(ii) $\Delta^{\alpha/2} q = 0$,

then the datum q is reproduced: $[q,u]_{x_i} = y_q$. If Q_i is D-invariant, the error is the linear functional

$$[q,u]_{x_i} - y_q = [\Delta^{\alpha/2} q, \tau]_{x_i}, \qquad q \in Q_i. \tag{7}$$

Proof. In (6), $[q,p_i^*]_{x_i} = \sum_s c_s \lambda_{i,s}(p_i^*) = y_q$. By (i), every $\partial^\gamma q$ (including $\gamma = 0$) lies in Q_i, and $\nu(x) \in N_i$, so $[\partial^\gamma q, \nu(x)]_{x_i} = 0$ for all $x \in U$: the middle term vanishes. This proves (7) when Q_i is D-invariant, and the last term vanishes by (ii). □

Condition (i) concerns the datum and the data at its own node only; it holds for every datum when Q_i is D-invariant, the classical setting of multivariate Hermite interpolation [**TODO:** cite de Boor–Ron, Sauer]. Condition (ii) holds whenever $\deg q < \alpha$, but also, for instance, for every harmonic symbol when $\alpha = 2$. By (7) the reproduced data at a D-invariant node form the subspace $\{q \in Q_i : \Delta^{\alpha/2} q = 0\}$ plus the kernel of the functional $\big[\cdot, \tau\big]_{x_i}$ on $\Delta^{\alpha/2} Q_i$; for instance, at $\alpha = 2$ the errors in $\partial_x^2 f$ and $\partial_y^2 f$ coincide, $\partial_x^2 - \partial_y^2$ is reproduced, and the error in $\partial_x^3 f$ is three times that in $\partial_x \partial_y^2 f$.

Corollary 4.5 (Hermite data)**.** *Let x_i carry classical Hermite data of order $k \le m$, i.e. $Q_i = \operatorname{span}\{\xi^\beta : |\beta| \le k\}$. Then every $q \in Q_i$ with $\Delta^{\alpha/2} q = 0$ is reproduced. In particular:*

- *(a) the derivative datum $D^\beta f(x_i)$ is reproduced whenever $\sum_j \lfloor \beta_j/2 \rfloor < \alpha/2$;*
- *(b) all data at x_i are reproduced if $\alpha > k$, i.e. for $\alpha = k+1$ when k is odd and $\alpha = k+2$ when k is even;*
- *(c) for $\alpha = 2$ the monomial data covered by (a) are exactly the multilinear ones ($\beta_j \le 1$ for all j, e.g. $\partial_x \partial_y$, $\partial_x \partial_y \partial_z$), whatever their order.*

Proof. Q_i is D-invariant, so Corollary 4.4 applies; (a) and (c) are Lemma 3.3 with $\ell = \alpha/2$, and (b) follows from $\deg q \le k < \alpha$. □

Corollary 4.5(b) proves the conjecture of [1, §3] that $\alpha = 2k$ suffices for data up to order k (they also observed that $\alpha = 2$ fails and $\alpha = 4$ works for $k = 2$); for $k \ge 3$ it is a strict improvement, e.g. $\alpha = 4$ suffices for $k = 3$. The next result shows that condition (ii) cannot be dropped, and with it that (b) is sharp (Corollary 4.7).

Proposition 4.6 (Necessity of (ii), leading order)**.** *In addition to Assumption 3.5, assume that the data at the nodes other than x_i are unisolvent on Π_m by themselves. Let $q \in Q_i$ satisfy (i), with $\deg q = \alpha$ and $\Delta^{\alpha/2} q \ne 0$. Then, for every choice of the data at x_i, the datum q is reproduced only if the data at the other nodes lie in a fixed affine hyperplane.*

Proof. By (i) the middle term of (6) vanishes, and $\Delta^{\alpha/2} q = c$ is a non-zero constant, so the error is $c\,\tau(x_i) = c\,\sigma(x_i)(x_i) = c\,s_0(x_i)$ with s_0 from the proof of Theorem 4.1. The error is an affine function of the data at the other nodes (with y_i fixed); it vanishes for the data $y_j = \Lambda_j p_i^*$, so it suffices to exhibit data for which it does not.

Since $\deg q = \alpha \ge 2$, some derivative $\partial^\gamma q$ with $|\gamma| = \alpha$ is a non-zero constant, so $1 \in Q_i$ by (i): point evaluation $\ell(p) = p(x_i)$ is a datum at x_i. It vanishes on N_i and is non-zero on Π_m, hence non-zero on R.

Choose $y_j = \Lambda_j(p_i^* + v)$, $j \ne i$, for some $v \in \Pi_m$. Then $B_{x_i}(p) = \langle K(p - p_i^* - v), p - p_i^* - v\rangle$ with $K = \sum_{j \ne i} \frac{w_j(x_i)}{\rho_i(x_i)} \Lambda_j^* \Lambda_j$, which is positive definite on Π_m by the additional assumption. The minimiser n_0 over N_i is characterised by $K(n_0 - v) \perp N_i$, i.e. $K(n_0 - v) \in R$, so $\nabla_R B_{x_i}(p_i^* + n_0) = 2K(n_0 - v)$ and $s_0 = (L^* L)^{-1} K w$ with $w = v - n_0$. As v ranges over Π_m, w ranges over the K-orthogonal complement $W = \{w : Kw \in R\}$ of N_i, so Kw ranges over all of R and s_0 over all of R. Choosing v with $\ell(s_0) \ne 0$ completes the proof. □

Corollary 4.7 (Sharpness)**.** *Under the hypotheses of Proposition 4.6 on the other nodes, let x_i carry classical Hermite data of order $k \le m$. If $\alpha \le k$, the datum $\partial_1^\alpha f(x_i)$ is not reproduced for data at the other nodes outside an affine hyperplane. Hence $\alpha > k$ is the exact condition for all Hermite data of order k to be reproduced for all data values.*

Proof. $q = \xi_1^\alpha \in Q_i$ satisfies (i) by D-invariance, $\deg q = \alpha$ and $\Delta^{\alpha/2} \xi_1^\alpha = \alpha! \ne 0$. □

4.4. **The role of condition (i).** When (i) fails, the middle term of (6) involves derivatives of ν, which are governed by how the weights of the *other* nodes vary near x_i. Two situations can be described completely.

Theorem 4.8 (Frozen minimiser)**.** *Let Assumption 3.5 hold and suppose that the frozen minimiser Z of (5) is constant near x_i. Then every $q \in Q_i$ with $\deg q \le \alpha$ satisfies*

$$[q,u]_{x_i} - y_q = \big[\Delta^{\alpha/2} q, \partial_t n(x_i,0) + \tau\big]_{x_i}, \tag{8}$$

where $\Delta^{\alpha/2} q$ is a constant. In particular every $q \in Q_i$ with $\deg q < \alpha$, or with $\deg q = \alpha$ and $\Delta^{\alpha/2} q = 0$, is reproduced, whether or not (i) holds.

Proof. By Theorem 4.1, $\nu(x) = n(x,t(x))$ with $n(x,0) = Z(x)$ constant, so $\nu(x) = Z(x_i) + t(x)\,g(x)$ with $g(x) = \int_0^1 \partial_t n(x,\theta t(x))\,d\theta$ smooth and $g(x_i) = \partial_t n(x_i,0)$. Since t vanishes at x_i to order α, $D^\gamma[t\,g](x_i) = 0$ for $|\gamma| < \alpha$ and $D^\gamma[t\,g](x_i) = (D^\gamma t)(x_i)\,g(x_i)$ for $|\gamma| = \alpha$. In the middle term of (6): the $\gamma = 0$ term is $\big[q, \nu(x_i)\big]_{x_i} = 0$ because $q \in Q_i$ and $\nu(x_i) \in N_i$; the terms with $0 < |\gamma| < \alpha$ vanish because the constant $Z(x_i)$ is killed by D^γ and $t\,g$ has vanishing derivatives of these orders; the terms with $|\gamma| > \alpha$ vanish because $\partial^\gamma q = 0$; and the terms with $|\gamma| = \alpha$, in which $\partial^\gamma q$ is a constant, add up by Lemma 3.4 to

$$\sum_{|\gamma|=\alpha} \frac{1}{\gamma!} (D^\gamma t)(x_i)\,\big[\partial^\gamma q, g(x_i)\big]_{x_i} = \big[\Delta^{\alpha/2} q, \partial_t n(x_i,0)\big]_{x_i}.$$

Together with $\big[q, p_i^*\big]_{x_i} = y_q$ and the last term of (6) this gives (8). □

Corollary 4.9. *The frozen minimiser is constant near x_i, and hence Theorem 4.8 applies, in each of the following cases:*

- *(a)* single scale: *$w_j(x)/\rho_i(x) = c(x)\kappa_j$ near x_i for all $j \ne i$, with a positive function c and constants $\kappa_j > 0$; this includes the case of a single other node ($K = 2$) and weights whose germs at x_i are proportional;*
- *(b)* interpolatory restriction: *the functionals $\lambda_{j,s}$, $j \ne i$, that do not vanish identically on N_i restrict to a basis of the dual of N_i (so that the frozen problem can fit them exactly whatever the weights).*

Proof. (a) $B_x = c(x)\hat{B}$ with $\hat{B}$ independent of x, and the minimiser of $c(x)\hat{B}$ over $p_i^* + N_i$ does not depend on $c(x) > 0$. (b) The frozen objective is a positively weighted sum of squared residuals of functionals that are either constant on $p_i^* + N_i$ or, by assumption, can be made zero simultaneously by exactly one $n \in N_i$, which is then the minimiser for every choice of positive weights. □

Example 4.10. These cases are not exotic: for two nodes, gradient data at x_i without a function value are reproduced for $\alpha \ge 2$; in one dimension with $m = 1$, the data $f'(0)$, $f(1)$, $f'(5/2)$ are all reproduced (case (b): only $f(1)$ sees the constants, which form N_i at the slope nodes). Adding a further value datum destroys both properties. [**TODO:** numerical confirmation in Section 7.]

In general the frozen minimiser moves, and a failure of (i) at first order produces an explicit error.

Proposition 4.11 (First-order error)**.** *Let Assumption 3.5 hold and let $q \in Q_i$ with $\deg q < \alpha$, such that $\partial^\gamma q \in Q_i$ for all $|\gamma| \ge 2$. Then*

$$[q,u]_{x_i} - y_q = \sum_{j=1}^{d} \big[\partial_j q, \partial_{x_j} Z(x_i)\big]_{x_i}, \qquad \partial_{x_j} Z(x_i) = -H^{-1}\,\partial_{x_j}\big[\nabla_N B_x\big](p_i^* + Z(x_i))\Big|_{x=x_i}, \tag{9}$$

where H is the Hessian of B_{x_i} restricted to N_i.

Proof. In (6) the last term vanishes because $\deg q < \alpha$, and the middle terms with $\gamma = 0$ or $|\gamma| \geq 2$ vanish as in Corollary 4.4. For $|\gamma| = 1$, $D_j\big[\partial_j q, \nu(x)\big]_{x_i} = \big[\partial_j q, \partial_{x_j}\nu(x_i)\big]_{x_i}$ and $\partial_{x_j}\nu(x_i) = \partial_{x_j} n(x_i, 0) + \partial_t n(x_i, 0)\,\partial_j t(x_i) = \partial_{x_j} Z(x_i)$ since $\alpha \geq 2$. Differentiating $\nabla_N B_x(p_i^* + Z(x)) = 0$ gives the formula for $\partial_{x_j} Z$. □

The right-hand side of (9) is an explicit linear functional of the data; for pure power weights ($\rho \equiv 1$) it is a rational function of the node positions. It vanishes in the cases of Corollary 4.9 and for symmetric configurations (e.g. $\partial_{x_1} w_j(x_i) = 0$ for all j), but it is non-zero in the explicit examples of Section 5, and therefore, by analyticity, for node positions in an open dense set for those data patterns.

Conjecture 4.12. *Let $\rho \equiv 1$, let Q_i be D-invariant and the other nodes unisolvent by themselves. If $\Delta^{\alpha/2} q \neq 0$ then, for node positions outside a proper algebraic subset, q is not reproduced for data outside a hyperplane.*

Remark 4.13. Proposition 4.6 proves the conjecture for $\deg q = \alpha$. It is supported by the experiments of Section 7 (e.g. $\partial_x^2 \partial_y f$ at $\alpha = 2$ fails even for $K = 2$). Characterising the node configurations for which a failure of (i) causes an error, beyond the first-order formula (9), is open.

5. Derivative data without function values

5.1. **Counterexamples.** Komargodski and Levin [1, §3] extend their Hermite MLS, with weights $|x - z|^{-\alpha}$, $\alpha \in 2\mathbb{N}$, and local spaces containing the constants and the linear functions, to directional derivatives prescribed at points that need not carry function values, and state that the derivative data are interpolated, the proof being "straightforward … by slight modifications" of the co-located case. Condition (i) fails for such data: the symbol of a first derivative differentiates to the constant 1, which is not in Q_i when no value is prescribed at the node. By Theorem 4.8 this does not always cause an error, but it can, and does in the following examples, computed in exact rational arithmetic (scripts in `code/proofs`).

Proposition 5.1. *Let $d = 1$, $m = 1$ (local polynomials $a + by$), $\alpha = 2$, $\rho_i \equiv 1$, and prescribe $f(0) = f(2) = 0$ and $f'(1) = f'(3) = 1$. Then*

$$u(x) = \frac{x\,(x-1)(x-2)\,(x^2 - 4x + 5)}{2x^4 - 14x^3 + 37x^2 - 42x + 19},$$

which satisfies $u(0) = u(2) = 0$ but $u'(1) = -1$ and $u'(3) = 31/25$.

Proof. With $w_z(x) = (x - z)^{-2}$ the normal equations of (3) for (a, b) are

$$\begin{pmatrix} w_0 + w_2 & 2w_2 \\ 2w_2 & 4w_2 + w_1 + w_3 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ w_1 + w_3 \end{pmatrix},$$

whose solution gives $u = a + bx$ as stated; the denominator has no real zeros. The remaining values follow by direct computation. □

Proposition 5.2. *Let $d = 2$, $m = 1$, $\alpha = 2$, $\rho_i \equiv 1$, and prescribe $f = 0$ at $(0, 0)$, $(2, 0)$, $(0, 2)$ and $\nabla f(1, 1) = (1, 1)$. Then u is a rational function with a positive denominator, $u = 0$ at the three value points, and $\nabla u(1, 1) = (5/9, 5/9)$.*

Proof. Exact computation as in Proposition 5.1 (three unknowns); the denominator is a quartic with leading form $35(x_1^2 + x_2^2)^2$ whose only real critical point has value $11.21\ldots > 0$. [**TODO:** display u in an appendix.] □

The failure persists for $\alpha = 4$ and for $m = 2$ (Section 7).

5.2. **Completion.** Given data at x_i with symbol space Q_i, let $\overline{Q}_i$ be the smallest D-invariant space containing Q_i; it has the same maximal degree as Q_i, since differentiation lowers degrees. Choose a basis $\tilde{q}_{i,1}, \dots, \tilde{q}_{i,r}$ of a complement of Q_i in $\overline{Q}_i$.

Proposition 5.3 (Completion)**.** *Augment the data at x_i by the functionals $\tilde{q}_{i,l}(D)f(x_i)$ with arbitrary values $\tilde{y}_{i,l}$. If Assumption 3.5 holds for the original data, it holds for the augmented data, and every original datum $q \in Q_i$ with $\Delta^{\alpha/2}q = 0$ (in particular every datum of order less than α) is reproduced by the augmented interpolant, whatever the values $\tilde{y}_{i,l}$.*

Proof. The augmented symbols at x_i form a basis of $\overline{Q}_i$, so (A2) holds; (A3) is preserved by adding data. The prescribed value y_q of an original datum is unchanged, because its representation in the augmented basis is unique. Every $q \in Q_i \subset \overline{Q}_i$ satisfies (i) with respect to the D-invariant space $\overline{Q}_i$, and Corollary 4.4 applies. □

The values $\tilde{y}_{i,l}$ influence the accuracy, not the exactness. In the experiments they are estimated by a smooth (non-singular) moving least-squares fit of the given data.

6. Numerical evaluation

[**TODO:** centred, scaled basis; log-normalised weights; SVD least squares; the two-block reformulation near a node; closed-form node value of Theorem 4.1; comparison with ε-regularised normal equations (Figure: e5_stability).]

7. Numerical experiments

[**TODO:** exactness experiments (verification method: high-precision stencils at two radii); Birkhoff patterns; the degenerate configurations of Corollary 4.9 and symmetric layouts; the error identities (7); completion; convergence (state only what the parameter study supports).]

8. Conclusions

[**TODO:** summary; open problems: Conjecture 4.12; configurations in which (i) matters; convergence orders; better estimators for completion.]

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